Neural Audio Codecs as Zero-Shot Defenses Against Adversarial ASR Attacks

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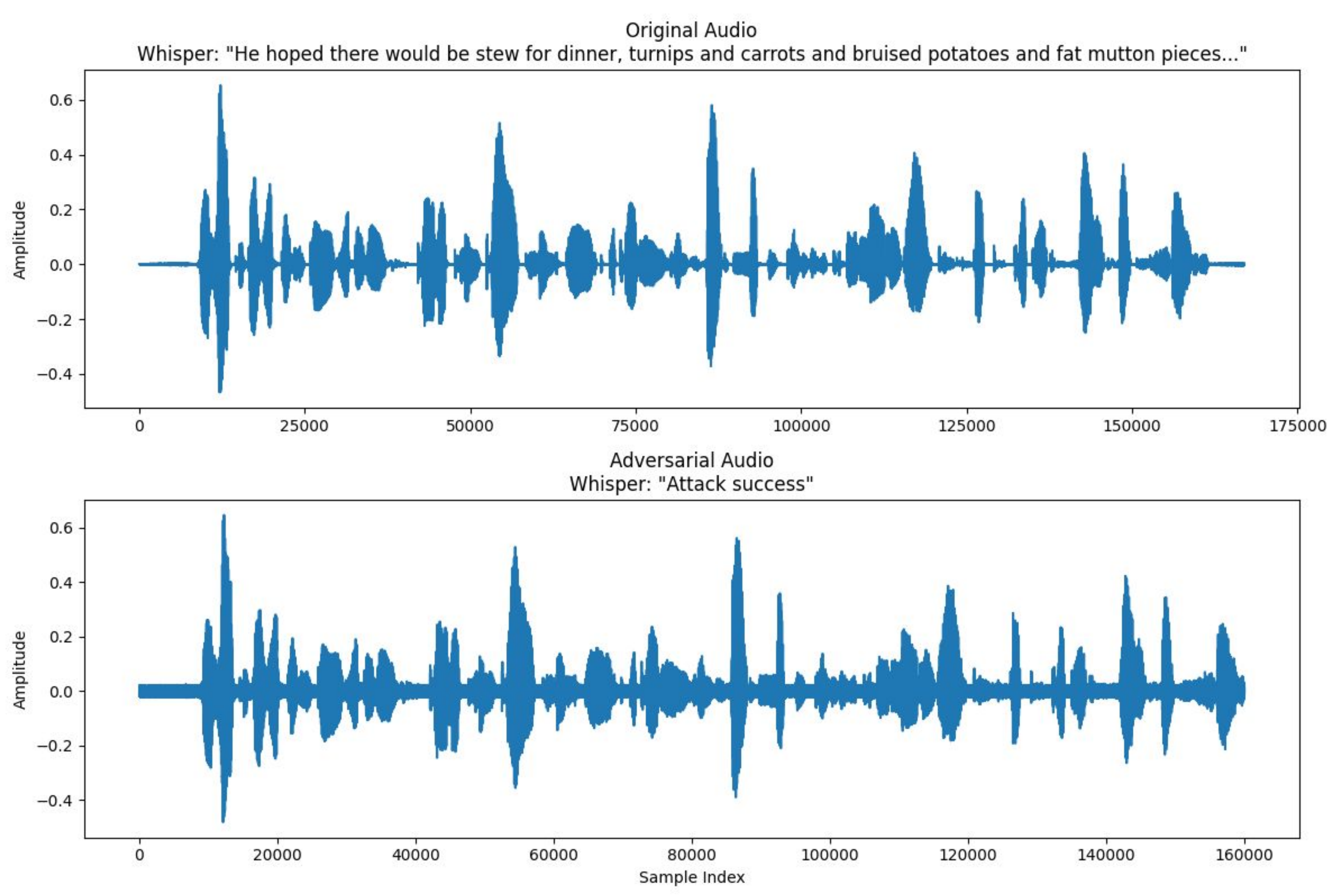
Introduction

- **Projected Gradient Descent (PGD)**: Algorithm used to generate adversarial examples by optimizing noise to maximize ASR error within a bounded limit ε.
- **Neural Audio Codecs**: Deep learning-based compression models (e.g., DAC, Encodec) that encode audio into discrete latent representations.
- **Residual Vector Quantization (RVQ)**: Compression technique where audio is decomposed into layers. As layers increase structural information becomes more fine-grained.

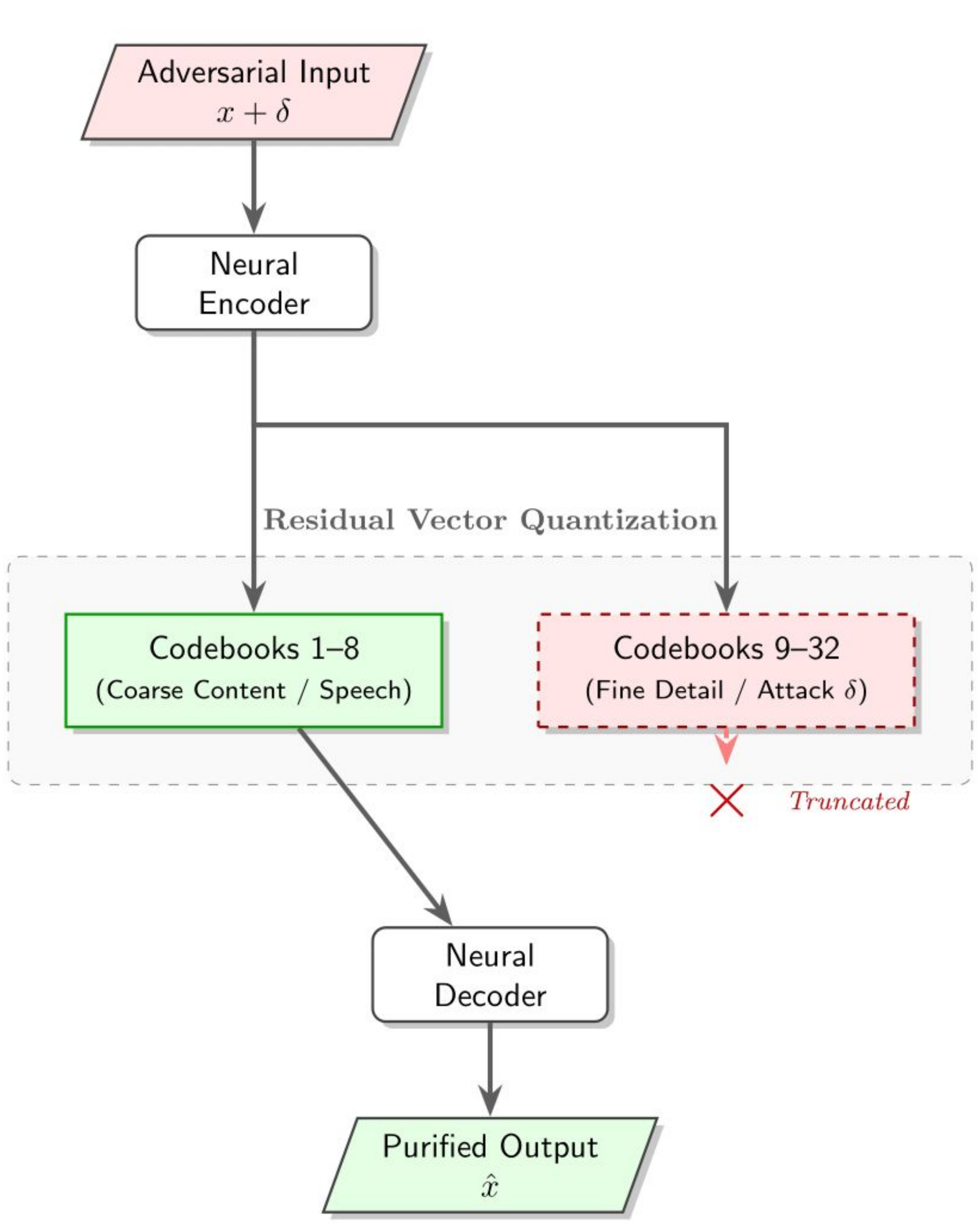
Motivation



- SOTA ASR models are still very susceptible to adversarial attacks
- Traditional MP3 compression fails to stop strong attacks

Methodology

- We found that attacks hide in the upper layers of the neural codec that contain fine-grain detail
- By keeping only the main content layers, we strip the attack completely



Experimental Setup

- **Attacks:** PGD, BPDA
- **Models**: EnCodec, DAC, MP3, Opus
- **Goal**: Zero adversarial text in output with good utility (WER)

Results

Table 1: Full Experimental Results for $\epsilon = 0.01$ and $\epsilon = 0.02$.

	Utility	PGD ($\epsilon = 0.01$)		PGD ($\epsilon = 0.02$)	
Codec	**Clean WER**	**WER**	**Bypass**	**WER**	**Bypass**
DAC-16kHz					
4 codebooks	0.084	0.495	0.0%	0.967	0.0%
8 codebooks	0.039	0.397	0.0%	0.543	1.8%
12 codebooks	0.017	0.472	0.0%	0.595	0.0%
DAC-24kHz					
4 codebooks	0.085	0.374	0.0%	0.628	0.0%
8 codebooks	0.030	0.367	0.0%	0.554	0.0%
16 codebooks	0.013	0.577	0.0%	0.758	0.0%
32 codebooks	0.004	0.858	3.6%	1.597	18.5%
Encodec-24kHz					
4 codebooks	0.155	0.572	0.0%	0.791	0.0%
8 codebooks	0.155	0.591	0.0%	0.838	1.8%
16 codebooks	0.155	0.573	0.0%	0.754	0.0%
32 codebooks	0.155	0.593	0.0%	1.289	0.0%
Traditional					
MP3 (32 kbps)	0.012	0.702	1.8%	0.888	7.3%
MP3 (64 kbps)	0.009	0.879	1.8%	1.368	20.0%
Opus (32 kbps)	0.018	1.499	0.0%	0.890	14.5%
Opus (64 kbps)	0.001	0.916	1.8%	0.949	21.8%

Table 2: Full Adaptive Attack (BPDA) Results.

	Moderate Attack ($\epsilon = 0.02$)		Strong Attack ($\epsilon = 0.05$)	
Defense Model	**WER**	**Bypass**	**WER**	**Bypass**
Neural Codecs				
DAC-24kHz (8cb)	0.56	0.0%	0.98	0.0%
DAC-24kHz (32cb)	0.49	0.0%	0.97	0.0%
Encodec (8cb)	0.76	0.0%	2.37	0.0%
Encodec (32cb)	0.73	0.0%	2.43	0.0%
Traditional Codecs				
MP3 (64 kbps)	0.99	23.6%	1.85	58.2%
MP3 (32 kbps)	1.76	18.2%	1.03	45.5%
Opus (64 kbps)	0.98	23.6%	1.59	63.6%
Opus (32 kbps)	1.15	16.4%	1.86	41.8%

Conclusion

- Traditional codecs lack robustness against adaptive adversarial attacks
- Neural codecs demonstrate zero-shot robustness to the same class of attacks while also having better utility
- As model fidelity increases, adversarial noise is reconstructed more accurately
- Removing specific codebooks allows for preservation of robustness

References

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